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Ecosystem Consequences of Cyanobacteria in the Northern Baltic Sea

Cyanobacteria of the Baltic Sea have multiple effects on organisms that influence the food chain dynamics on several trophic levels. Cyanobacteria contain several bioactive compounds, such as alkaloids, peptides, and lipopolysaccharides. A group of nonribosomally produced oligopeptides, namely microcystins and nodularin, are tumor promoters and cause oxidative stress in the affected cells. Zooplankton graze on cyanobacteria, and when ingested, the hepatotoxins (nodularin) decrease the egg production of, for example, copepods. However, the observed effects are very variable, because many crustaceans are tolerant to nodularin and because cyanobacteria may complement the diet of grazers in small amounts. Cyanobacterial toxins are transferred through the food web from one trophic level to another. The transfer rate is relatively low in the pelagic food web, but reduced feeding and growth rates of fish larvae have been observed. In the benthic food web, especially in blue mussels, nodularin concentrations are high, and benthic feeding juvenile flounders have been observed to disappear from bloom areas. In the littoral ecosystem, gammarids have shown increased mortality and weakening of reproductive success under cyanobacterial exposure. In contrast, mysid shrimps seem to be tolerant to cyanobacterial exposure. In fish larvae, detoxication of nodularin poses a metabolic cost that is reflected as decreased growth and condition, which may increase their susceptibility to predation. Cyanobacterial filaments and aggregates also interfere with both hydromechanical and visual feeding of planktivores. The feeding appendages of mysid shrimps may clog, and the filaments interfere with prey detection of pike larvae. On the other hand, a cyanobacterial bloom may provide a refuge for both zooplankton and small fish. As the decaying bloom also provides an ample source of organic carbon and nutrients for the organisms of the microbial loop, the zooplankton species capable of selective feeding may thrive in bloom conditions. Cyanobacteria also compete for nutrients with other primary producers and change the nitrogen (N) : phosphorus (P) balance of their environment by their N-fixation. Further, the bioactive compounds of cyanobacteria directly influence other primary producers, favoring cyanobacteria, chlorophytes, dinoflagellates, and nanoflagellates and inhibiting cryptophytes. As the selective grazers also shift the grazing pressure on other species than cyanobacteria, changes in the structure and functioning of the Baltic Sea communities and ecosystems are likely to occur during the cyanobacterial bloom season.

INTRODUCTION

Nitrogen-fixing cyanobacteria have occurred for at least 7000 years in the Baltic Sea, according to sediment core analyses (1). Both their frequency and intensity have, however, been increasing during the last few decades (2–4), probably because of anthropogenic eutrophication, which creates favorable nutrient conditions for cyanobacteria through internal loading of phosphorus from anoxic sediments (5). At the same time, the ratio of toxic *vs.* nontoxic cyanobacterial species has increased in late summer (6).

Cyanobacterial blooms in the Baltic Sea are usually dominated by *Nodularia spumigena*, *Aphanizomenon flos-aquae,* and *Anabaena* spp. These taxa are able to fix molecular nitrogen (7), store phosphorus (8), and adjust their vertical position in the water column by their gas vesicles (9). Although cyanobacterial blooms are typical for the summer, there are cyanobacteria present in the water column all year round, as well as in the sea ice (10, 11). The blooms usually first occur in the open sea, but may drift or be formed in the archipelagoes, enclosed bays, and in the vicinity of littoral zones. In addition, certain cyanobacteria do not contain flotation devices and are therefore abundant on surfaces in benthic and littoral habitats.

In the Baltic Sea cyanobacterial blooms occur abundantly in all subbasins of the central Baltic Sea and in the Gulf of Finland. In the Gulf of Bothnia, especially Bothnian Bay, the nutrient regime is less suitable for cyanobacteria dominance because the nitrogen (N) : phosphorus (P) ratio is relatively higher because of the large freshwater runoff containing humic substances, which efficiently precipitate phosphorus (12). During the last decades, however, cyanobacterial blooms have been observed more frequently in the Bothnian Sea area as well (2, 13). In the southernmost Baltic Sea, nitrogen fixation becomes limited by iron (14) and sulfate (15), and therefore cyanobacterial mass occurrences are generally not formed in Kattegat (16).

Because of their affinity to relatively high water temperatures and a stable water column, cyanobacteria are most abundant during late summer (17, 18). Initiation of a cyanobacterial mass occurrence depends on nutrients, temperature, and hydrographic conditions. *N. spumigena* is mainly dependent on phosphorus reserves and a suitable temperature, whereas *A. flos-aquae* may occur in cooler water but is relatively more dependent on wind-induced mixing and upwelling of nutrients (18). Because the blooms are formed and accumulate in the surface layer, a large part of cyanobacterial biomass is decomposed in the water column (19). On the other hand, the occurrence of cyanobacterial pigments in the sediments and high concentrations of nodularin determined in the surface of sediments after the bloom season, suggest that some part of the cyanobacterial bloom sediments and reaches the seafloor (4).

To sum up, cyanobacteria occur and form blooms everywhere in the Baltic, excluding Bothnian Bay. Seasonally they occur throughout the year, but the largest blooms are most common in the open sea, in the surface layer, and in summer. Consequently, the most significant ecosystem effects of cyanobacteria can also be expected in the pelagic food web in summer. When the cyanobacterial bloom drifts ashore, or reaches the bottom sediments, the littoral and benthic ecosystems may be similarly affected.

The mechanisms behind the formation and decay of cyanobacterial blooms in the Baltic Sea have been investigated for several decades (cf. 18). In contrast, the studies on the effects

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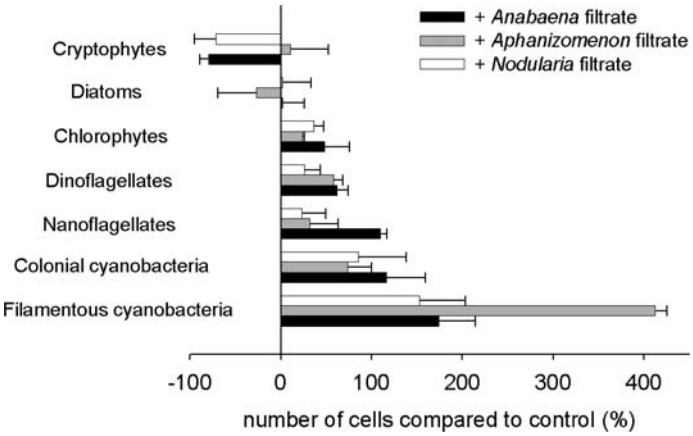

Figure 1. Allelopathic effects of cyanobacterial filtrates, taken from exponentially growing *Nodularia spumigena*, *Aphanizomenon* sp., and *Anabaena* sp. cultures, on a natural Baltic phytoplankton community after 4-day exposure compared with those of the control medium (with no cyanobacteria) (n = 3, mean + SD). The concentration of cyanobacteria in the experimental units corresponded ca. 75 μg l^{-1} chlorophyll *a* (39).

of cyanobacteria have a much shorter history (cf. 20–22). Since both proceeding eutrophication and climatic change in the Baltic Sea area may increase the duration and intensity of the cyanobacterial blooms, these studies are as relevant as those studying cyanobacteria bloom dynamics. In this article, we will review the current knowledge on the individual, population, and ecosystem level effects of Baltic cyanobacteria.

TYPES OF INTERACTION

Cyanobacteria may affect other organisms, and hence, food webs and ecosystems, in various ways. The direct toxic effects are well known from both freshwater and brackishwater studies (23). In most cases, toxic effects are observed after ingestion of cyanobacteria. Grazing on cyanobacteria does take place, but since most pelagic secondary producers, such as copepods, marine cladocerans, mysid shrimps, and fish larvae, are capable of selective feeding, they are not prone to be directly intoxicated. On the other hand, ingestion of cyanobacteria is not needed for harmful effects, because some dissolved nodularin is transferred to zooplankton directly from the water (24). Nodularin, its derivatives, and possibly also other bioactive compounds produced by cyanobacteria can also be transferred in the food web. Planktivores feeding on zooplankton can be affected especially, even though they would not encounter cyanobacteria themselves (25).

Cyanobacteria also influence the Baltic Sea ecosystem in various indirect ways. As for consumers, these include visual and mechanical interference of feeding on preferred food items, and changing the pelagic environment from a transparent one to one offering refuge for smaller prey. Cyanobacteria also obviously influence other primary producers by competing for nutrients (26) and by increasing the N : P ratio of the water because of their nitrogen-fixing ability (7). They may also directly harm the other algae by their bioactive compounds, i.e., allelopathy.

The toxic and chemical effects of cyanobacteria are called direct effects; indirect effects indicate those affecting one property of the target organism (such as behavior or feeding) or community, whereas the main consequence can be seen in another property (e.g., susceptibility to predation, or reproduction). All of these interspecific interactions give rise to the community- and ecosystem-level consequences of cyanobacteria, discussed at the end of this review.

TOXIC AND CHEMICAL EFFECTS OF CYANOBACTERIA

Cyanobacteria produce a variety of bioactive compounds. Some of these chemicals, like alkaloids, peptides, and lipopolysaccharides, are harmful to other organisms (27). The best known are the hepatotoxins, e.g., nodularin and microcystins, which are tumor promoters (28) and inhibit eukaryotic serine–threonine protein phosphatases PP1 and PP2A (29). These hepatotoxins expose the affected cells to oxidative stress, which possibly explains the observed effects (30, 31).

There are several studies about the factors controlling the nodularin production rates of *N. spumigena* (Baltic Sea studies, cf. 32), but the ultimate reason for nodularin production is not fully understood. Several explanations for algal toxin production have been suggested, such as toxins acting as grazing deterrents (33), allelochemicals (34), essential secondary metabolites (35), intracellular signalling compounds (36), or quorum-sensing chemicals (37), but a consensus is lacking. Notably, benthic cyanobacteria are known to produce cytotoxins, but no microcystins have so far been detected from them (38), which suggests that production of nodularin or microcystins in the Baltic Sea may, for some reason, only be advantageous in the pelagic environment.

Allelopathy

Toxic *N. spumigena* reduces the growth of certain phytoplankton species, either via allelochemichals (39) or by elevating the pH to a level that other species cannot tolerate (40). Purified nodularin produced by *N. spumigena* did not have any negative effect on the growth of cryptophyte *Rhodomonas* sp. (41), indicating that nodularin production has some other function than allelopathy. In contrast, cell-free filtrates of *N. spumigena*, *Aphanizomenon* sp., and *Anabaena* sp., which probably contain several bioactive compounds, stimulated the growth of certain phytoplankton groups, especially cyanobacteria, but also chlorophytes, dinoflagellates, and nanoflagellates, whereas they inhibited the growth of cryptophytes (Fig. 1) (39). Cyanobacteria can thus have both positive and negative effects on other phytoplankton groups.

Effects on Zooplankton

The effect of cyanobacteria on grazers in limnic environments has generally been considered to be negative because of the toxins they produce (42, 43), but also because cyanobacteria are difficult to manage because of their colonial form (44) and because they have a low nutritional value (45). They also contain protease inhibitors, which in turn affect the ability of zooplankton to digest cyanobacteria (46).

In the Baltic Sea, the peak abundances of crustacean zooplankton and cyanobacteria overlap in summer, suggesting that many zooplankton species have to cope with cyanobacteria in various ways. Experimental studies have shown that a diet consisting solely of toxic (or even nontoxic) *Nodularia* can increase mortality and decrease the egg production rate (Fig. 2) in Baltic copepods (47), indicating that their nutritional value is poor. In the field, cyanobacterial community is consumed at variable rates (48–50), and there is also experimental evidence that toxic *N. spumigena* can be eaten by copepods (51, 52). Experiments have shown that copepods can avoid feeding on toxic strains, which leads to a preferential consumption of nontoxic cyanobacteria, whereas other species can reject filamentous cyanobacteria completely from their diet (Fig. 3) (53). On the other hand, aging and degradation of cyanobacterial filaments make them more suitable food for zooplankton (49, 54). In addition, cyanobacterial aggregates are associated

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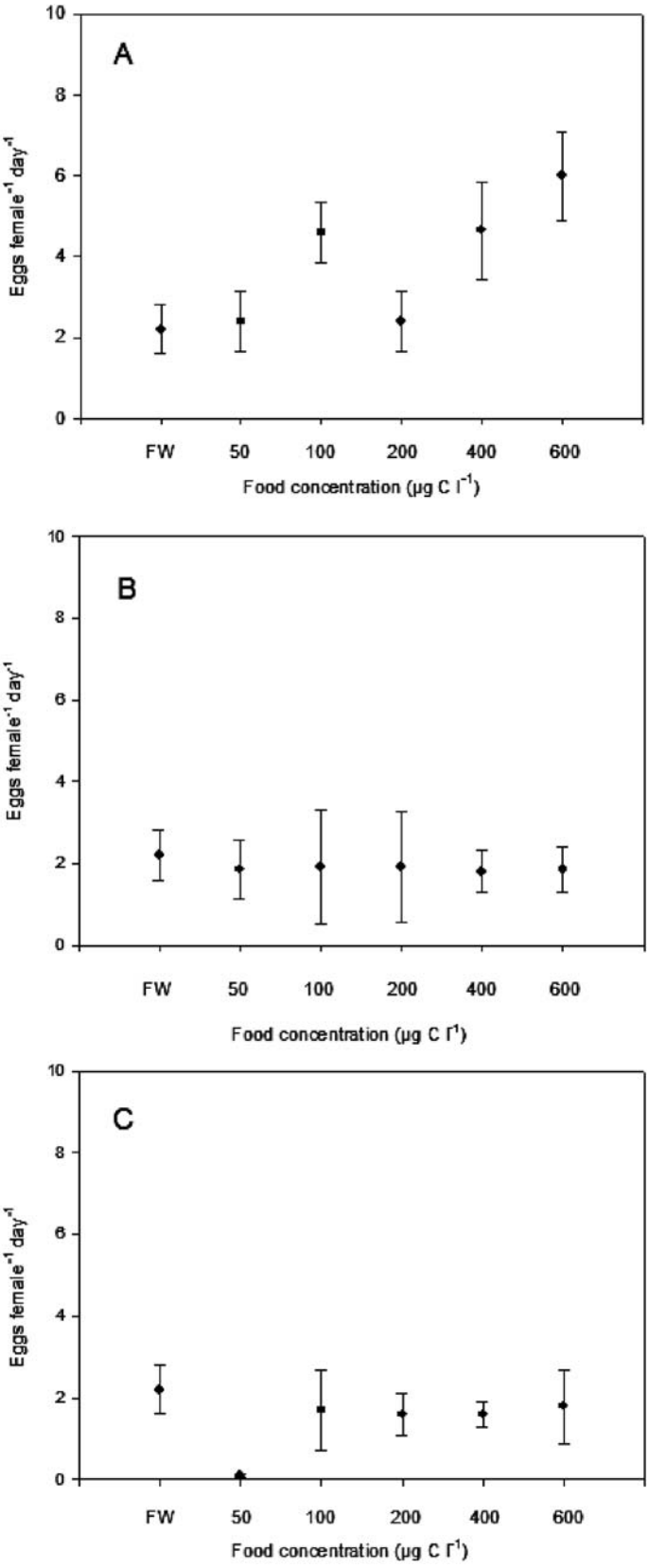

Figure 2. *Eurytemora affinis*. Egg production rates (mean ± SE) in different concentrations (μg C L^{-1}) of (A) *Brachiomonas submarina*, (B) nontoxic *Nodularia* sp., and (C) toxic *Nodularia* sp. (47).

with other phytoplankton groups, such as diatoms, bacteria, flagellates, and ciliates (55), which are suitable food items for mesozooplankton (51, 56, 57). Poor quality of cyanobacteria food can be thus supplemented by other, more nutritious, food items (Fig. 4) (58).

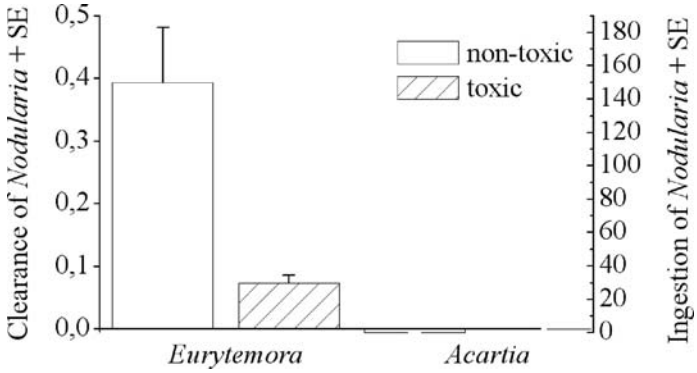

Figure 3. Clearance (mL ind h^{-1}) and ingestion rates (ng C ind^{-1} h^{-1}) of copepods *Eurytemora affinis* and *Acartia bifilosa* (mean ± SE) on nontoxic and toxic *Nodularia* sp. provided at a concentration of 400 μg C L^{-1}; n = 12 (53).

There are, however, only a few field observations on the production and growth of zooplankton within a cyanobacterial bloom. Recent field experiments indicated low egg production rates in copepods living within a cyanobacterial bloom, even with good food added (Karjalainen et al., unpubl. data). This suggests that during the cyanobacterial bloom season copepods may be food limited and that additional food is not primarily converted to egg production, but to somatic growth or maintenance.

Effects on Other Baltic Invertebrates

Feeding rate of mysid shrimps decreased when they were fed with zooplankton in the presence of nontoxic *Nodularia* filaments (Fig. 5) (59). However, mysid shrimps seem to be rather tolerant to *N. spumigena* containing nodularin, as their survival was not affected by toxic cyanobacteria during several weeks of exposure (59). In addition, no negative effect was observed on their growth when fed with zooplankton exposed to cyanobacterial extract from toxic *N. spumigena* for a two-week period (25).

Littoral invertebrates, such as amphipods, isopods, and molluscs, are exposed to cyanobacteria when a surface scum is

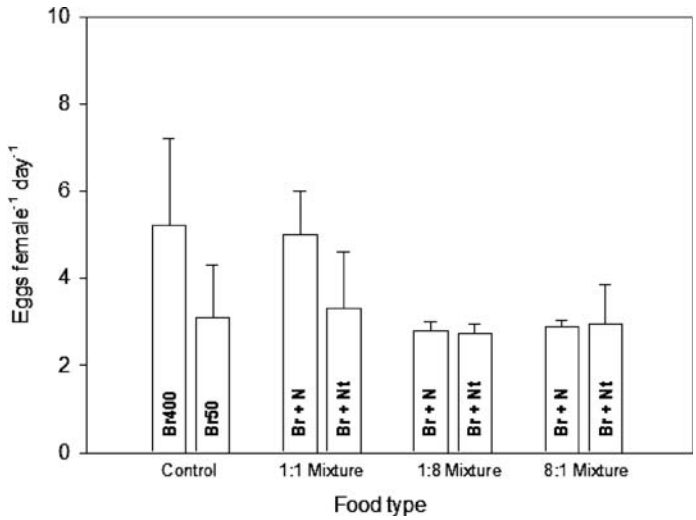

Figure 4. *Eurytemora affinis* egg production rates (mean + SE) in mixture of *Brachiomonas submarina* (Br) and nontoxic and toxic *Nodularia spumigena* (N and Nt, respectively), and with corresponding concentrations (400 μg C L^{-1} and 50 μg C L^{-1}) of *B. submarina* as sole food (Br400 and Br50, respectively). Ratios (1 : 1), (8 : 1), and (1 : 8) refer to food ratios of 200 + 200 μg C L^{-1}, 400 + 50 μg C L^{-1}, and 50 + 400 μg C L^{-1}, of *B. submarina* and *Nodularia* sp., respectively (47).

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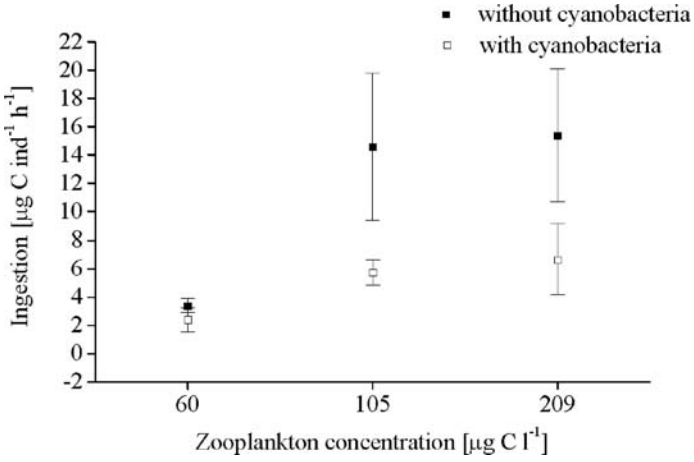

Figure 5. Ingestion rates (mean ± SE) of adult mysids (*Mysis mixta*) on three concentrations of *Acartia bifilosa* in the absence or presence of cyanobacteria (400 μg C L^{-1}) (59).

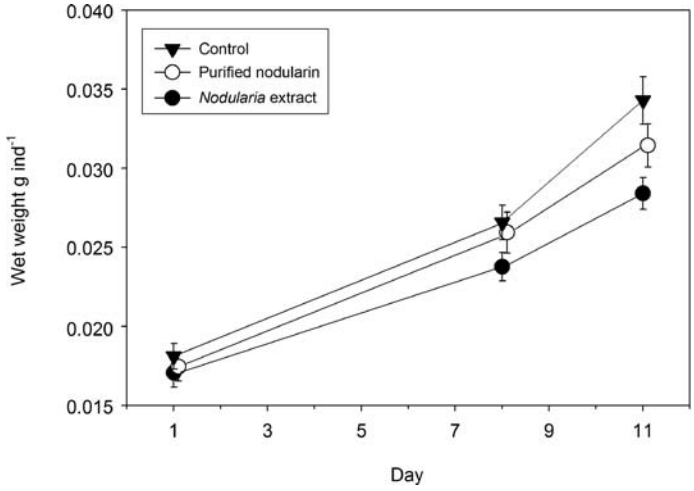

Figure 7. Growth of pike larvae measured as wet weight (mean ± SE) during an 11-day experiment. Zooplankton pretreatments were filtered seawater (control), purified nodularin (20 μg L^{-1}), and *Nodularia spumigena* extract (mean nodularin concentration ± SE: 18.5 ± 2.3 μg L^{-1}) (25).

drifted ashore. Gammarids, albeit mainly feeding on macroalgae, have been shown to be affected by toxic cyanobacteria: their mortality increased, and their reproductive success decreased during heavy *N. spumigena* exposure (Fig. 6) (60).

Benthic organisms encounter cyanobacteria and their toxins as well, because high concentrations of nodularin have been determined in the surface of sediments after the cyanobacterial bloom season (61). Suspension feeders, such as mussels, are especially prone to ingesting sedimented cyanobacterial filaments and their toxins. The nodularin concentrations in blue mussels are one of the highest that have been measured in Baltic Sea organisms (62–65). Effects of cyanobacteria on growth or reproduction of blue mussels or benthic organisms inhabiting the soft sediment have not been investigated, but there are indications that their physiological condition is affected by toxic *N. spumigena* (66).

Effects on Fish

Fish seem to be more sensitive to cyanobacteria than invertebrates. If toxic *N. spumigena* cells are ingested, they can cause severe liver damage to fish (67). Especially starved fish are generally more vulnerable to cyanobacterial hepatotoxins, because starvation decreases protein phosphatase activity and glycogen reserves of fish (68). Hepatotoxic microcystins, closely related compounds to nodularin, can damage the gill structures of fish, exposing them to other dissolved toxins in the water (69), and increase the drinking rate of fish, further accelerating the toxin uptake from the water (70). The oxidative stress caused by microcystins can also affect fish at the cellular level (31).

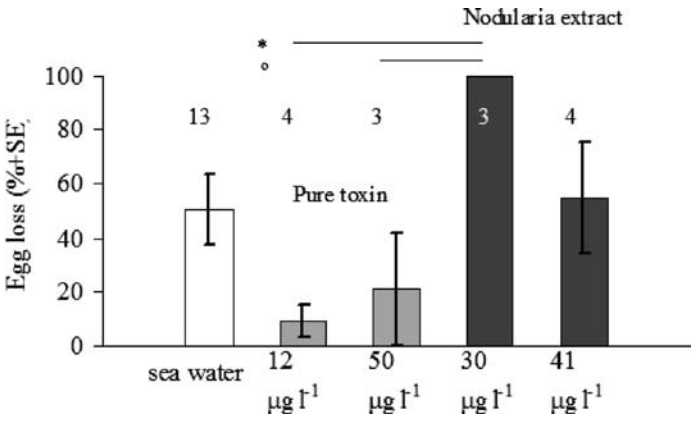

Figure 6. *Gammarus zaddachi*. Abortion of eggs (mean ± SE) from the female brood pouch. Multiple comparison results are shown by asterisks ($p < 0.05$) and circles ($p < 0.1$). Initial nodularin concentrations, measured from carbon concentrations, are shown on the x-axis. Replicate number is shown above the bars. White bar = seawater treatment, light grey bars = purified nodularin treatment, dark grey bars = *Nodularia spumigena* treatment (60).

Nodularin, and probably also other bioactive compounds, can be transferred to fish via their zooplankton food (Fig. 7). In an experimental set-up where pike larvae were fed on cyanobacteria-exposed zooplankton, their feeding and growth rates were lower than when feeding on nonexposed zooplankton (25). Notably, this took place without the fish encountering the cyanobacteria themselves.

The negative effects on fish can also be caused by the low nutritional quality of cyanobacteria. Cyanobacteria lack essential fatty acids DHA (22:6ω3) and EPA (20:5ω3), which are required for the normal growth, development, and reproduction of fish and zooplankton (45, 71). Thus, cyanobacteria-fed zooplankton can also be poor food for planktivorous fish because of their low content of essential fatty acids.

It is important for juvenile fish to grow rapidly to decrease the risk of predation (72, 73). On the other hand, detoxication of microcystins decreases the oxidative stress in exposed fish embryos, but energy is allocated simultaneously in detoxication processes (74). Similarly, detoxication of nodularin is an energy-demanding process, and if young fish must cope with ingested toxins, their growth rates may decrease (e.g., 25). Exposure to cyanobacterial toxins from *N. spumigena* blooms may also affect the condition of fish. During a few weeks exposure to *N. spumigena*, the growth of exposed juvenile three-spined sticklebacks was reduced and growth became allometric, i.e., their length increased more rapidly than their weight. The RNA/DNA ratio, which correlates strongly with growth rate (75), did not increase in exposed fish (Fig. 8) (Pääkkönen et al., unpubl. data), although food consumption increased.

The benthic feeding fish, such as gobies, sand-eels, cod, and flounders carry the highest amounts of nodularin in their tissues, especially liver (61, 62). After the extensive cyanobacterial bloom near the Estonian north coast in the summer 2002, the 1-year-old flounder almost completely disappeared from experimental catches. This suggests that, since the young flounders mainly feed in shallow waters, they may be more strongly affected by the toxic cyanobacterial blooms than the

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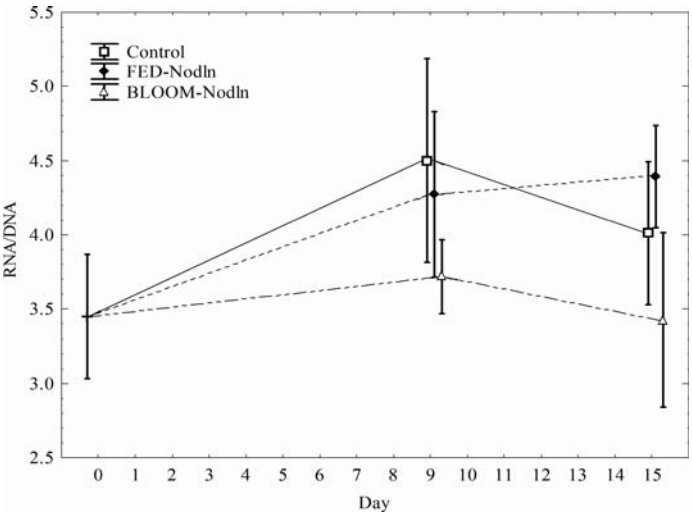

Figure 8. The RNA/DNA ratio (mean ± SE) of juvenile three-spined stickleback after 9 and 15 days of exposure to toxic cyanobacteria *Nodularia spumigena*. No change in RNA/DNA ratio was observed in fish that ate exposed zooplankton in bloom conditions (BLOOM-nodln) as compared to control or fish exposed cyanobacterial toxin through diet (FED-nodln). Source: Pääkkönen et al., unpubl. data.

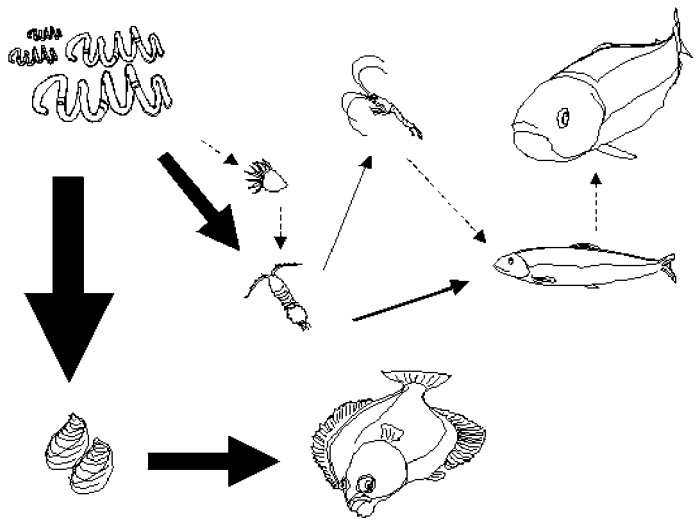

Figure 9. A schematic presentation of the fate of the nodularin in the pelagic and benthic food webs. The thickness of arrows depicts the differences in transfer rates between species or compartments (22).

older ones that returned into the Estonian coasts after the bloom had disappeared (76).

THE FATE OF NODULARIN IN BALTIC SEA ECOSYSTEM

Microcystins and nodularin are rather stable compounds, resisting heat, sunlight, and microwaves (77–79). Photodegradation of microcystins can occur in the surface layer of water bodies, where humic substances or algal pigments are present (80), but mainly they need to be biologically broken down. Specialized microbes are capable of metabolizing microcystins (81, 82), but also other organisms can detoxify them by using glutahione-S-transferase (83), and transform the original molecule to less toxic conjugates. The conjugates of nodularin have been found in several Baltic Sea organisms (84). This biological degradation of toxic molecules demands energy from other vital functions. The trophic transfer of nodularin seems to take place in the Baltic Sea, but the transfer efficiency from one trophic level to the next is relatively low, since <1% is transferred from one level to another (22). Notably, in the pelagic food web the concentrations within organisms are lower than in the benthic communities (Fig. 9).

The majority of nodularin is found in *N. spumigena* cells during cyanobacterial blooms. Since nodularin is an intracellular toxin, nodularin is mainly released to the surrounding water during bloom decay (27). Depending on the stratification and mixing conditions, this dissolved nodularin is then diluted to the surrounding water. Nodularin may also adsorb to clay particles similarly to microcystins and sediment from the surface layer to the bottom (85). In the open sea, the concentrations of dissolved nodularin are usually very low (86, 87), whereas in littoral and shallow areas (88, 89) the concentrations of dissolved nodularin can be high, even several days after the cyanobacterial filaments have disappeared from the water.

The highest concentrations of nodularin in Baltic Sea organisms have been found in species that are directly in contact with cyanobacterial filaments, such as bottom-dwelling molluscs that are exposed to the decaying cyanobacteria aggregates that are sedimenting from the euphotic zone (62). In the pelagic food web, except the grazing zooplankton, concentrations are low (Fig. 9). Because toxin degradation takes place at every trophic level, no biomagnification, but rather biodilution of nodularin takes place in the food web (22).

INDIRECT EFFECTS OF CYANOBACTERIA

Effects on Behavior

In addition to cellular damage, it has been demonstrated that microcystins can also affect the behavior of both aquatic invertebrates (90–92) and vertebrates (93, 94). These behavioral changes can then have indirect consequences on the organisms, e.g., by lowering their feeding, reproduction, or survival (cf. 58, for effects of *Prymnesium patelliferum* on the behavior of *Eurytemora affinis*). Effects on survival are, however, difficult to predict,because changes in behavior, such as increasing passiveness, may increase or decrease the predation rates by weakening the escape reactions of zooplankton and by decreasing prey conspicuousness to predators. Since microcystins and nodularin are very similar in their mode of action, it is possible that also nodularin could have an effect on the behavior of aquatic organisms.

Mechanical and Visual Interference

Cyanobacterial filaments increase the water turbidity and make the prey search of visual predators, such as planktivorous fish, more difficult (95). Consistently, the feeding rate of pike larvae decreased in experimental conditions where *N. spumigena* filaments were present, compared to a situation without filaments (Fig. 10) (96).

Mechanical interference by cyanobacterial filaments may also hinder the feeding of various grazers and predators, because cyanobacterial filaments and aggregates can attach to their feeding appendages. Clogging of appendages of mysid shrimps (*Mysis mixta*) was observed in experimental conditions (59), which suggests that their feeding rates may be lower in the bloom than outside it. However, it is not known whether this phenomenon occurs in the field, because the mysid shrimps are able to avoid the surface bloom by vertical migration.

The Refuge Hypothesis

The visual and mechanical interference by cyanobacteria aggregates also implies that a cyanobacteria bloom may provide a refuge for small prey. If the predators for various reasons (avoidance of toxins, clogging of appendages, poor prey

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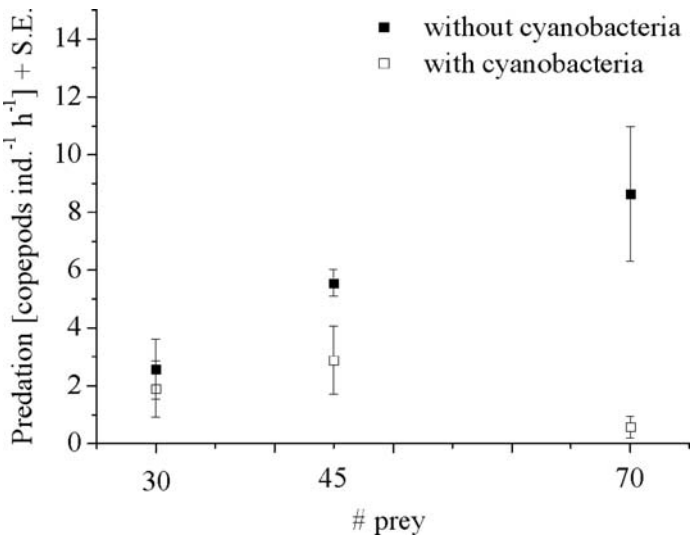

Figure 10. Feeding experiments. Pike larvae predation rate (mean ± SE) on copepods *Acartia bifilosa* at three different concentrations (30, 45, 70 ind. L^{-1}) in the absence or presence of a constant concentration of nontoxic cyanobacteria *Nodularia sphaerocarpa* (800 μg DW L^{-1}) (96).

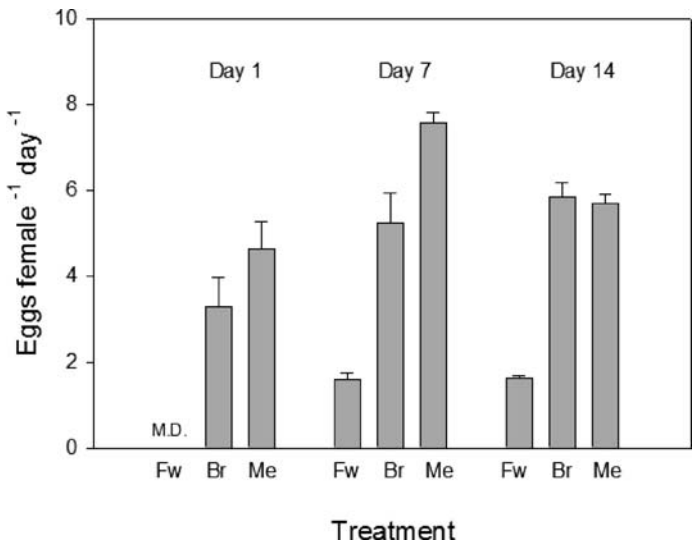

Figure 11. Egg production of *Acartia bifilosa* in filtered seawater (Fw), in *Brachiomonas submarina* control (Br), and in mesocosm water (Me) containing a cyanobacteria bloom, at experiment days 1, 7, and 14 from the start of the mesocosm experiment (mean + SE). M.D. = missing data (51).

detection success) avoid the bloom, the prey that can cope with the cyanobacteria may find both food and refuge inside the bloom. In other sea areas, such as Peel Harvey estuary, Australia (97), there are observations about fish avoiding areas of dense *N. spumigena* blooms, but in a survey done in the Gulf of Finland, planktivorous clupeids did not avoid areas with surface floating blooms. Instead, the fish biomass correlated positively with cyanobacterial toxins in the water (Pääkkönen et al., unpubl. data), which suggests that the fish rather were attracted by the bloom than avoided it.

COMMUNITY EFFECTS OF CYANOBACTERIA

In the Baltic Sea, the pelagic ecosystem in summertime is mainly based on the microbial loop (bacteria–flagellates–ciliates–copepods) (e.g., 98), instead of the shorter diatom–copepod-based grazing chain that prevails in many other sea areas. The decomposing cyanobacterial bloom, being a rich source of bacteria, heterotrophic flagellates, and ciliates (56), can sustain high copepod growth and reproduction (Fig. 11) (51). Furthermore, the selectively feeding zooplankton can, by feeding preferentially on the nontoxic phytoplankton, indirectly enhance the growth of toxic algae (cf. 99–101).

Since cyanobacteria can form massive blooms that overwhelm other pelagic species, they may influence the structure of the whole pelagic ecosystem. Long-term effects on grazing zooplankton can be expected, if cyanobacterial mass occurrences last for a longer period of time. However, individuals and populations differ in their sensitivity to cyanobacteria and their toxins (102, 103). Previous feeding history on toxic cyanobacteria affects the sensitivity of daphnids to cyanobacterial exposure (104, 105) and, even within one species, populations inhabiting different geographical areas may show different responses to algal toxins (106).

Community-level changes in fish populations may be possible as well. Decreased feeding, condition, and growth of fish larvae probably affect their survival rates, potentially leading to variations in year-class strength. These changes can be postulated to decrease the top-down control of planktivores on zooplankton, which may further enhance the importance of the bottom-up effects and processes taking place within the actual bloom.

CONCLUSIONS

Increasing cyanobacterial blooms increase the quantity of available food for consumers, but at the same time may decrease the relationship between high- and low-quality food items. The largest effects are expected on the species inhabiting the same environments and seasons as cyanobacteria, i.e., the surface waters in summer, and the bottom, to where the cyanobacterial filaments sediment. On the other hand, cyanobacteria usually occur very patchily. During intense open sea blooms, coastal areas may be virtually cyanobacteria-free, and the upwelling-induced coastal blooms may be restricted to limited areas. Also in the bloom region, dense accumulations are usually intervened by stretches with much lower concentrations of cyanobacteria. Therefore motile animals, such as vertically migrating zooplankton, mysid shrimps, and fish, can either avoid bloom areas or be attracted by them. In the case of fish, the avoidance is in turn counteracted by the fact that planktivores may obtain toxins through their food, even without encountering a bloom.

Increasing cyanobacterial blooms probably shift the pelagic community toward one favored by a microbial loop, i.e., high growth of flagellates, ciliates, and selectively feeding copepods. Motile predators and those with a high tolerance for toxins will be favored, whereas surface-dwelling, herbivorous, and nonselective species with limited motility will be disfavored.

Not all of the observed effects of cyanobacteria are because of their toxicity or poor nutritional value. Various indirect effects, such as effects on behavior and energy allocation, as well as effects on other primary producers, may induce and amplify the effects of cyanobacteria on several trophic levels. Although less than 1% of the ingested nodularin is usually transferred to the next trophic level, even relatively small doses of toxins may affect the organisms. The effects of a long-term exposure to cyanobacteria are currently virtually unknown. In freshwater ecosystems, lake-wide changes in the genotypic variance toward better resistance against cyanobacterial toxins have been observed (in daphnids) (107). It is not known if such adaptation occurs, or has already occurred, in the Baltic Sea.

To sum up, cyanobacterial blooms are an integral part of the Baltic Sea summer ecosystem. Direct negative effects of cyanobacteria have mostly been observed in laboratory

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conditions, whereas few clearly negative effects have been verified in field surveys. This reflects the complexity of biotic interactions in natural conditions, but may also prove that most Baltic organisms are able to cope with, or avoid, the negative effects of cyanobacterial toxins and filaments in their natural environment. On the other hand, the fact that cyanobacteria do have sublethal effects on the feeding, growth, and reproduction of various organisms suggests that cyanobacteria can influence the structure and dynamics of the ecosystem of the Baltic Sea, especially if the intensity and duration of cyanobacteria blooms will increase along with the proceeding eutrophication.

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